

Depth-Anything V3 (DA3) Validation Report

This report summarizes the evaluation of Depth-Anything V3 (DA3) depth estimation against ground-truth (GT) data in a Gazebo simulation environment.

Experiment Setup:

- Drone moves forward toward a wall
- Fixed colored markers placed on wall
- GT depth computed from 3D geometry (camera frame Z)
- DA3 depth sampled at detected marker positions

Key Results:

- Mean DA3/GT ratio: ~ 1.45
- Standard deviation: ~ 0.27
- Mean absolute error: ~ 1.31 meters

Interpretation:

- DA3 systematically overestimates distance
- Error increases with distance (non-linear distortion)
- Behavior is consistent across all markers (no color bias)
- System is stable (low jitter), indicating reliable measurements

Model Relationship:

$$GT \approx 0.50 * DA3 + 0.61$$

This indicates:

- Scale error (DA3 approximately twice actual depth)
- Bias offset present
- Non-linear behavior across depth range

Conclusion:

- DA3 provides consistent but biased depth estimates
- Suitable for calibration and correction
- Current system pipeline (projection, detection, synchronization) is correct

Recommendation:

Proceed to model calibration (linear or non-linear correction) using collected data.